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Handled by
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Our Reference
PC-EP-21145437

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European Patent Application No 22725665.8
in the name of Lutron Technology Company LLC

Dear Mesdames/Sirs,

In reply to the Communication pursuant to Article 94(3) EPC amended claims 1-17, as full replacement for the claims presently on file, are respectfully submitted together with amended description page 1. Our observations on the objections raised in the Communication are presented below.

AMENDMENTS

The claims have been amended to include reference signs to increase their intelligibility, as requested by the Examining Division. The description has also been amended to acknowledge prior art document D1.

OBSERVATIONS

Re Basis (Item 2 of the communication - Rule 137(4) EPC)

The Communication requests that any amendments to the claims be clearly identified. Accordingly, a table is provided below mapping the features of the current claims to the application as originally filed.

Claim	Basis in the application as filed	Comment
Claim 1	Original Claim 1	Identical
Claim 2	Original Claim 2	Identical
Claim 3	Original Claim 3	Identical

Claim 4	Original Claim 4	Identical
Claim 5	Original Claim 5	Identical
Claim 6	Original Claims 6 and 7	Combined from original claims 6 and 7.
Claim 7	Original Claims 8 and 9	Combined from original claims 8 and 9.
Claim 8	Original Claim 10	Original claim 10 was renumbered to amended claim 8.
Claim 9	Original Claim 11	Original claim 11 was renumbered to amended claim 9.
Claim 10	Original Claim 12	Original claim 12 was renumbered to amended claim 10.
Claim 11	Original Claims 13 and 14	Combined from original claims 13 and 14.
Claim 12	Original Claim 15	Original claim 15 was renumbered to amended claim 12.
Claim 13	Original Claim 16	Original claim 16 was renumbered to amended claim 13.
Claim 14	Original Claim 17	Original claim 17 was renumbered to amended claim 14.
Claim 15	Original Claims 18 and 19	Combined from original claims 18 and 19.
Claim 16	Original Claim 20	Original claim 20 was renumbered to amended claim 16.
Claim 17	Original Claim 21	Original claim 21 was renumbered to amended claim 17.

Re Inventive Step (Item 3 of the communication - Article 56 EPC)

The Communication maintains an objection of lack of inventive step, based on the reasoning in the IPRP which considers document **D1 (JP 2010 230492 A)** as the closest prior art.

While we agree that D1 represents a suitable starting point, we respectfully submit that the claimed subject-matter is not obvious for the reasons set out below.

1. Closest Prior Art D1 and the Analysis in the IPRP

D1 discloses a PIR sensing circuit comprising a first amplifier stage (3a) and a second amplifier stage (4). D1 acknowledges a startup delay problem caused by the charging of a capacitor. Specifically, paragraph [0058] of D1 states that "a capacitor for passing the AC component" is installed "at the output of the primary filter 3a". The charging of this unlabelled series coupling capacitor causes the output of the subsequent amplifier stage (4) to be unstable at power-on.

To solve this, D1 proposes activating a "third sensitivity correction means 8". With reference to Figure 2 of D1, this means (8) is part of the feedback loop of the *first* amplifier stage (3a). By activating means (8), the time constant of this first stage is reduced, causing its output to stabilize more quickly, which in turn indirectly accelerates the charging of the downstream series coupling capacitor "at the output of the primary filter 3a".

The IPRP (section 2.1.1) focuses on the charging of capacitor C1 in D1. While C1 is involved in the mechanism, it is part of the filter loop that is manipulated, not the actual capacitor whose charging causes the delay problem (as discussed above, according to paragraph [0058] in D1, the problem in D1 stems from the unlabelled series capacitor). That is, the solution is an **indirect, open-loop** manipulation of a preceding amplifier stage.

2. Distinguishing Features

The control device of claim 1 differs from D1 in several fundamental aspects:

(i) **Different Target Capacitor and Function:** The capacitor being charged in claim 1 (the "first capacitor") is part of a filter circuit for the **second amplifier stage (340)**. This is distinct from D1, where the "problematic" capacitor is a series coupling capacitor between stages.

(ii) **Different Charging Mechanism:** Claim 1 defines a **direct** charging mechanism, wherein the charging current is conducted **"through the control circuit" (360)** to the target capacitor. D1 teaches an **indirect** mechanism, where an earlier amplifier stage is manipulated to affect the charging of a downstream component.

(iii) **Different Control Method:** The claimed control circuit (360) implements a **closed-loop feedback system**. It is configured to measure the capacitor voltage and **"stop charging...when the magnitude of a voltage...is greater than or equal to a threshold voltage"**. D1 is entirely silent on such a feedback mechanism; its process is an open-loop or timed procedure.

That is, the claimed direct charging mechanism operates as a closed-loop feedback system. As defined by the claim, the control circuit actively supervises the process by measuring the "magnitude of a voltage developed across the first capacitor" and comparing this real-time measurement against a defined "threshold voltage". The charging process is then terminated only when this threshold is reached, ensuring the capacitor is charged to a precise and known operational state. This stands in stark contrast to the open-loop process in D1, which provides no mechanism for measuring the resulting capacitor voltage to determine when to cease its startup procedure. The claimed invention's use of a closed-loop system is therefore a fundamentally different and more advanced technical approach.

3. Technical Effect and Objective Technical Problem

As discussed above, the claimed feedback mechanism ensures that the capacitor is charged to the exact required operational voltage, making the startup process robust against component tolerances and temperature variations. D1's indirect, open-loop method is inherently imprecise. Further, the claimed method stops charging the moment the target is reached, ensuring the device is operational in the minimum time necessary.

In view of these differences, a technical effect achieved by the claimed invention is to provide a **more precise, efficient, and reliable startup procedure** for a PIR sensor. An objective technical problem solved by the claimed invention may hence be formulated as **"how to provide a more precise, efficient, and reliable startup procedure for the PIR sensor in D1?"**

4. Non-Obviousness

A person skilled in the art, starting out from D1, would not find any teaching in the prior art as a whole that **would** (not simply could, but would) have prompted him, faced with the above objective technical problem, to modify or adapt the circuit in D1 while taking account of that teaching, thereby arriving at something falling within the terms of the claims, and thus achieving what the invention achieves.

More specifically, the Communication's reasoning suggests a modification of D1, but this would require a radical redesign far beyond obvious steps. More specifically, the skilled person would have to:

1. Abandon D1's indirect charging philosophy and **invent a new, direct charging mechanism** from the control circuit.
2. Decide to apply this new mechanism to a **different capacitor** (a filter capacitor within the second stage) serving a different function.
3. Further **invent a closed-loop feedback control system** to monitor and terminate this new charging process, a concept for which D1 provides no hint whatsoever.

It is not reasonable to assume that the skilled person would take all these steps in an effort to solve the objective technical problem without hindsight of the claimed invention. We therefore trust that the subject-matter of claim 1, and by extension its dependent claims, involves an inventive step.

Re Formal Objections

The claims have been provided with reference signs, as requested.

D1 has been identified and is briefly discussed in the opening part of the description, as requested.

It is regarded as unsuitable to put independent claim 1 in a two-part form since the inventiveness resides in the combination of features disclosed, which combination is not known from the prior art. A reorganization of the claim would give a distorted picture of the invention. This conclusion is in line with the Guidelines for Examination F-IV, 2.3.

REQUESTS

In view of the above, favourable action based on the enclosed claims is hereby respectfully requested. In the event that the Examiner is not prepared to issue a communication in accordance with Rule 71(3) EPC, an additional opportunity to submit amendments and/or arguments under Article 94(3) EPC and Rule 71(1) EPC is respectfully requested. Alternatively, the Examiner is welcome to contact the undersigned to discuss the case during an informal telephone conversation. As an auxiliary measure only, oral proceedings are requested.

Yours faithfully,

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Encls. Amended claims 1-17, Amended description page 1